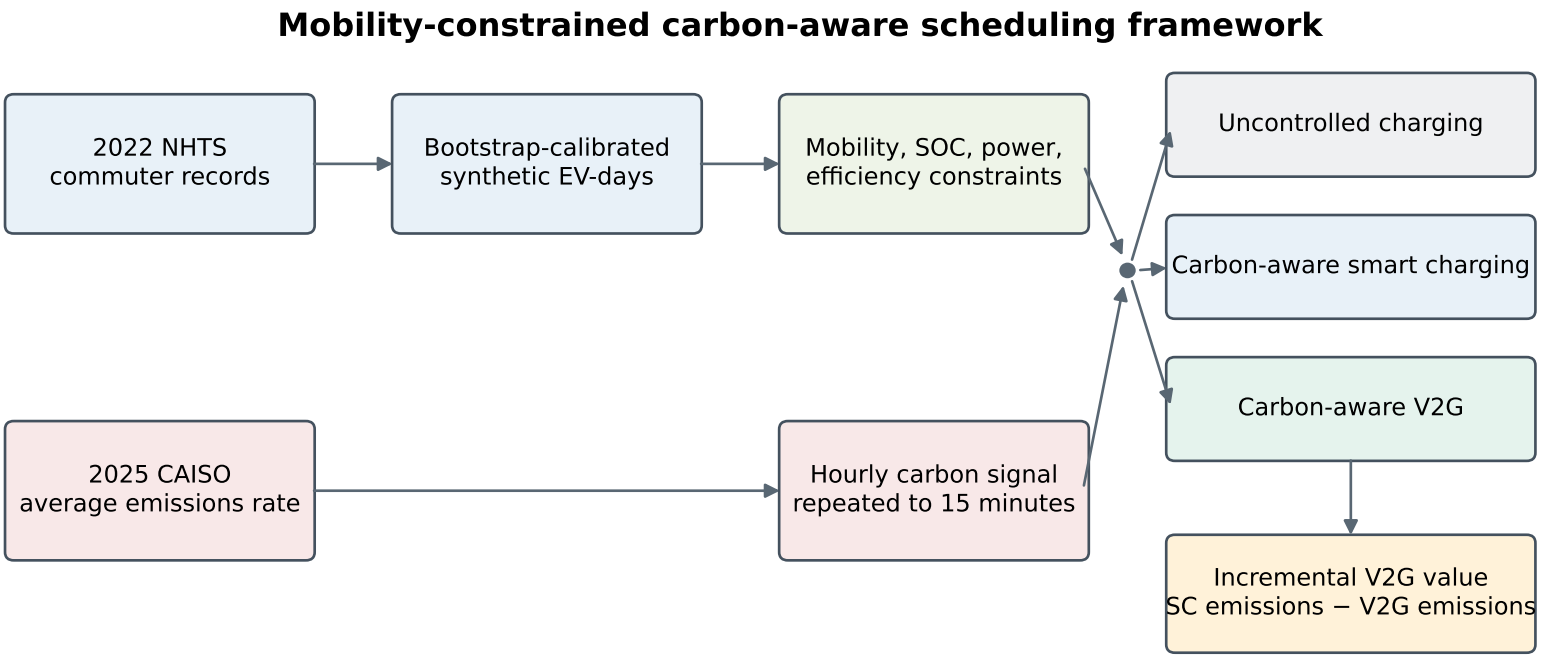
Mobility-constrained carbon-aware scheduling framework
2022 NHTS commuter records
Bootstrap-calibrated synthetic EV-days
Mobility, SOC, power, efficiency constraints
2025 CAISO average emissions rate
Hourly carbon signal repeated to 15 minutes
Uncontrolled charging
Carbon-aware smart charging
Carbon-aware V2G
Incremental V2G value
SC emissions − V2G emissions